Are you blinded by statistics?

1. Power of statistics

The oft-quoted statement that 97% of statistics are made up should show the general public that research does not discover facts, but serves other purposes. Researchers who fabricate statistics are rarely challenged since the assumption is that the data reported in research articles exists. This default setting of “honest and ethical researchers” is rather naïve, given the positive effect that publishing in top-ranked journals can have on academic careers.

2. Downtown

Large businesses are destroying the small town feel of our “downtown” area. Just last year, the number of large businesses in the city has increased by 75%.

(adapted from Brown & Keeley, 2011, p.171)

3. Big shoes, big test scores

Researchers administered an IQ test to a sample of a total 978 children. The possible grades on the test were poor, acceptable, good, and excellent. 65% of the children with larger shoes scored a "good" or better compared to only 35% among the children with smaller shoes. This result is highly statistically significant, the researchers said. Hopper proposed that shoe size has a positive effect on self-confidence, which translates into better learning and test-taking skills.

Steel, D. (31 May, 2001). Big shoes, big test scores. Carnegie Mellon University.
http://www.cmu.edu/CSR/case_studies/shoe_size.html

4. Nuclear power

There is no reason to worry about the new nuclear power plant being built in our city; the average amount of harm caused by nuclear accidents is rather low.

(adapted from Brown & Keeley, 2011, p.171)

5. GoFat

A new weight-loss drug, GoFat, is effective in helping obese people lose weight. In a clinical trial, only 6 out of 100 people on GoFat reported any side effects with taking the drug. The company manufacturing the drug argues “With 94% of people having positive results with GoFat, it is safe to say our pill is one of the most effective weight-loss pills in the market”.

(adapted from Brown & Keeley, 2011, p.171)

6. Water quality

The data (see Table 1 below) give a chi-square test statistic of 18.89 with 3 degrees of freedom and a p-value of 0.000288. Because this is a significant result, the local government needs to ascertain the reasons why tap water is less popular than the other choices.

Table 1. Data for the Chi-Square Goodness-of-Fit Test for First Preference in the Double-Blind Test

Type of Water	Tap	Aquafina	Fiji	Sam’s Choice	Total
Observed	12	27	44	26	109
Expected	27.25	27.25	27.25	27.25	109
Contribution to Test Statistic	8.534	0.002	10.296	0.057	18.889

7. Test scores

As can be seen in the table below, there is a substantial difference in the pre-test and post-test scores. In almost every instance subjects scores increased in the post test. The teaching strategy employed is a strong candidate to explain these significant increases.

Subject id	Pre test raw score	Post test raw score
1	52.5	53.5
2	37.5	45.0
3	22.5	24.0
4	20.0	22.5
5	47.5	46.0

8. Race and attendance

The following table shows the expected and observed frequencies of particular races attending a lecture on campus in spring 2015. The results are statistically significant, evidenced by the chi-square and p values. We can therefore conclude that the difference between expected and observed results did not occur by chance, and given the every action and state has a cause, the causality needs investigation. There are two likely candidates that can explain the larger than expected numbers, namely: the topic and the timing of the lecture.

	expected freq	observed freq	Row Totals
hispanic	20 (21.27) [0.08]	34 (32.73) [0.05]	54
asian	0 (12.12) [12.12]	20 (7.88) [18.65]	20
african-american	20 (16.15) [0.92]	21 (24.85) [0.60]	41
white	70 (84.70) [2.55]	145 (130.30) [1.66]	215
Column Totals	110	220	330 (Grand Total)

The chi-square statistic is 36.6159. The p-value is < 0.00001. This is significant at $p < 0.05$.

9. Bias at Berkley

The University of California, Berkley was sued for bias against women. Looking at the overall figures 11% fewer women were admitted than men (men, 44%: women, 35%). Yet, despite the fact that 1/3 more male applicants were admitted than female, the university showed that there was no bias.

Source: Bickel, P.J., Hammel, E.A. & O'Connell, J.W. (1975). Sex bias in graduate admissions: Data from Berkeley. Science, 187 (4175), 398–404